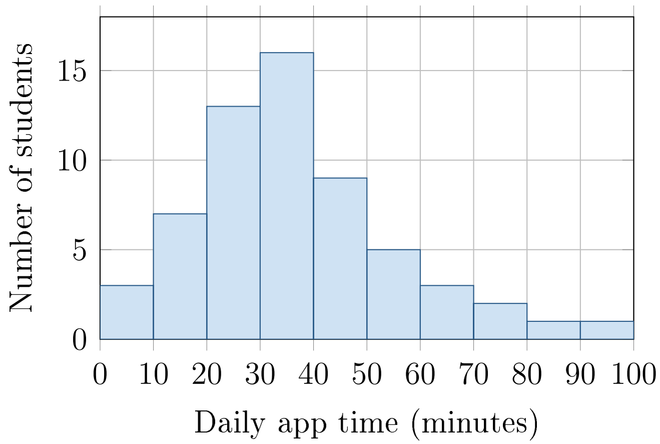
# Check the Model

Unit 2 · Topic 2.11 · THE PROBLEM

A school says daily app time is approximately normal with mean 40 minutes and standard deviation 12 minutes. The histogram shows 60 students; no recorded time was exactly 70 minutes. Table A gives a left area of 0.9938 for $z = 2.5$.

**Ben:** "About 16% should be above 52 minutes." **Ava:** "For 70 minutes, $z = 2.5$, so less than 1% should be above 70."




**FIRST TAKE — one sentence before you work the parts**

Do the times look balanced around 40 minutes, or does one side stretch farther? Point to the bars.

DOK 1 **(a)** State the proposed model's mean $\mu$ and standard deviation $\sigma$, with units.

DOK 2 **(b)** Check Ben's and Ava's model calculations. Then find the observed fraction of these 60 students above 70 minutes.

DOK 3 **(c)** Would you use Ben's and Ava's model results to describe these students? Use the histogram's shape and the observed versus predicted percent above 70. Explain what a reader could get wrong about long app times.

**Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.**